

Mathematical Foundations: DCGAN vs. WGAN-GP

A Comparative Analysis of Generative Adversarial Networks

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Abstract

This document describes the theoretical and mathematical foundations of Generative Adversarial Networks (GANs). We compare the standard Deep Convolutional GAN (DCGAN) regularized via Spectral Normalization against the Wasserstein GAN with Gradient Penalty (WGAN-GP). Furthermore, we formalize the Fréchet Inception Distance (FID) used for quantitative evaluation and analyze the computational overhead of these architectures.

1 Introduction

The traditional GAN setting defines the process of generating data as a minimax zero-sum game between a generator (G) and a discriminator (D) [1]. While typical GANs work to reduce the JS divergence between the actual data distribution $\mathbb{P}_r$ and the generated distribution $\mathbb{P}_g$, this is bound to cause the problem of gradient vanishing if the support of the two distributions do not overlap. Advanced network architectures overcome the above problems through an alternate objective function.

2 Evaluation Metric: Fréchet Inception Distance (FID)

To objectively evaluate the quality and diversity of synthesised images, we use the Fréchet Inception Distance (FID) [2]. In order to map real and generated images into a feature space continuously using a pre-trained Inception-v3 model, we can assume that the features follow a multivariate Gaussian distribution $\mathcal{N}(\mu_r, \Sigma_r)$ and $\mathcal{N}(\mu_g, \Sigma_g)$.

The FID is defined as the Fréchet distance (or Wasserstein-2 distance) between these two Gaussians:

$$\text{FID} = \|\mu_r - \mu_g\|_2^2 + \text{Tr}\left(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2}\right) \quad (1)$$

A lower FID score indicates that the generated distribution matches more closely the real data distribution.

3 Wasserstein-1 Distance and Kantorovich-Rubinstein Duality

Standard GANs often break down because the Jensen-Shannon divergence cannot measure the distance between distributions that do not overlap. To fix this, the Wasserstein GAN (WGAN) [3] uses the Wasserstein-1 distance, which measures the actual effort required to move mass from one distribution to another. Firstly, we are going to define the two distributions we are working with.

Let $\mathcal{X} = \mathbb{R}^{C \times H \times W}$ be the high-dimensional ambient space of our images.

- $\mathbb{P}_r$ is the **real data distribution**. In practice, real images do not fill the entire ambient space $\mathcal{X}$, but rather lie on a complex, lower-dimensional manifold within it.

- $\mathbb{P}_g$ is the **generated data distribution**. This is an implicit push-forward measure induced by the Generator network G , which maps low-dimensional latent noise vectors $z \sim p(z)$ (typically a standard Gaussian) into the image space $\mathcal{X}$.

The Wasserstein-1 distance quantifies the minimum "cost" required to transform the distribution $\mathbb{P}_g$ into $\mathbb{P}_r$. It is defined as:

$$W_1(\mathbb{P}_r, \mathbb{P}_g) = \inf_{\gamma \in \Pi(\mathbb{P}_r, \mathbb{P}_g)} \mathbb{E}_{(x,y) \sim \gamma} [\|x - y\|] \quad (2)$$

Here, $\Pi(\mathbb{P}_r, \mathbb{P}_g)$ denotes the set of all possible joint distributions $\gamma(x, y)$ whose marginals are exactly $\mathbb{P}_r$ and $\mathbb{P}_g$.

Because computing the infimum over all possible joint distributions is highly intractable in high-dimensional spaces, we apply the **Kantorovich-Rubinstein duality** [4]. This powerful theorem allows us to reformulate the metric from a minimization over transport plans to a maximization over a specific class of functions:

$$W_1(\mathbb{P}_r, \mathbb{P}_g) = \sup_{\|f\|_L \leq 1} \mathbb{E}_{x \sim \mathbb{P}_r} [f(x)] - \mathbb{E}_{\tilde{x} \sim \mathbb{P}_g} [f(\tilde{x})] \quad (3)$$

Rather than solving for the transportation problem directly, the dual formulation is searching for a function $f : \mathcal{X} \rightarrow \mathbb{R}$ that maximizes the difference between the average scores of the real and generated images. In the WGAN framework, a neural network referred to as the Critic learns this function. In order to guarantee the mathematical correctness, the condition of being a supremum imposes the strict requirement on the Critic of being **1-Lipschitz continuous** ($\|f\|_L \leq 1$).

4 Enforcing the Lipschitz Constraint

The fundamental engineering challenge of the WGAN formulation is practically enforcing the mathematical condition $\|f\|_L \leq 1$ on a deep neural network.

4.1 DCGAN and Spectral Normalization

The Lipschitz constraint is globally enforced in the DCGAN model through **Spectral Normalization** [5], which acts directly on the weights of the neural network. Spectral normalization guarantees that the Lipschitz constraint of a linear mapping $f(x) = Wx$ can be upper-bounded by the spectral norm of its weight matrix W , denoted by $\sigma(W)$.

Spectral normalization scales the weight matrix by dividing it by its spectral norm at each forward pass:

$$W_{SN} = \frac{W}{\sigma(W)} \quad \text{where} \quad \sigma(W) = \max_{h: h \neq 0} \frac{\|Wh\|_2}{\|h\|_2} \quad (4)$$

Normalizing each convolutional layer in the network guarantees that the operator norm of each layer is upper-bounded by one. Thus, due to the property of the 1-Lipschitz mappings, this technique restricts the whole discriminator network to be Lipschitz continuous without modifying the loss function.

4.2 WGAN-GP and Gradient Penalty

Alternatively, the WGAN-GP enforces the Lipschitz condition through a soft constraint dynamically applied during training. According to the properties of differentiable functions, a function is 1-Lipschitz almost everywhere if and only if the norm of its gradients is bounded by 1 ($\|\nabla_x f(x)\|_2 \leq 1$).

Instead of modifying the network weights, WGAN-GP adds a **Gradient Penalty** directly to the loss function to heavily penalize the model if the Critic’s gradient norm deviates from 1 [6]:

$$L_{Critic} = \underbrace{\mathbb{E}_{\tilde{x} \sim \mathbb{P}_g}[D(\tilde{x})] - \mathbb{E}_{x \sim \mathbb{P}_r}[D(x)]}_{\text{Wasserstein Loss}} + \underbrace{\lambda \mathbb{E}_{\hat{x} \sim \mathbb{P}_{\hat{x}}}[(\|\nabla_{\hat{x}} D(\hat{x})\|_2 - 1)^2]}_{\text{Gradient Penalty}} \quad (5)$$

where $\hat{x}$ are uniformly sampled points along straight lines interpolating between real points x and generated points $\tilde{x}$. By evaluating the gradient on these interpolated points, the penalty explicitly enforces the Lipschitz condition across the regions of the ambient space where the transport from $\mathbb{P}_g$ to $\mathbb{P}_r$ is actively occurring. λ is the penalty coefficient (typically set to 10 to balance the constraints).

5 Computational Complexity and Execution Time

Apart from convergence and stability, it is also crucial in Machine Learning to take into account the computational costs. Both models were trained and evaluated on a system equipped with an NVIDIA GeForce RTX 3060 GPU and an AMD Ryzen 5 processor.

The total recorded execution time was **2 hours and 10 minutes** for the DCGAN, and **2 hours and 15 minutes** for the WGAN-GP. Theoretically, the WGAN-GP is computationally heavier per epoch because the Critic is updated five times for every Generator update, and calculating the Gradient Penalty requires costly second-order derivatives.

However, the difference in execution time between the two is actually very small. This is because the repetitive computation of FID creates a huge bottleneck in computation time. Computing 2048-dimensional feature vectors using the pretrained Inception-v3 model is an extremely expensive process. Hence, the large amount of computation reduces the difference in purely training time.

6 Empirical Results and Convergence Stability

To validate the theoretical advantages of the Wasserstein-1 metric over the Jensen-Shannon divergence, both architectures were trained for 30 epochs on the CelebA dataset [7]. The primary quantitative comparison relies on the evolution of the Fréchet Inception Distance (FID).

- **DCGAN (Spectral Normalization):** While Spectral Normalization successfully prevented immediate mode collapse, the FID evolution displayed higher variance across epochs, characteristic of the JS divergence’s topological limitations.
- **WGAN-GP:** The training dynamics were markedly more stable. The continuous nature of the Wasserstein metric provided reliable gradients even in early epochs, resulting in a smoother, monotonic decrease in the FID score.

Qualitative Evaluation

While the quantitative metrics confirm the stability of the WGAN-GP framework, the qualitative visual synthesis of facial features provides the final empirical validation. For an interactive visual comparison of the generated images and latent space interpolations, please refer to the `Model_evaluation.ipynb` notebook provided in the repository.

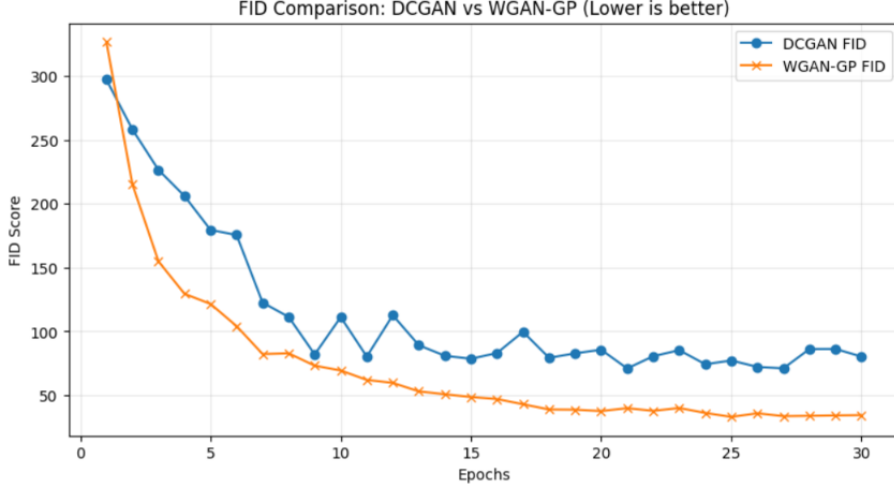


Figure 1: Evolution of the FID score over 30 epochs. WGAN-GP demonstrates smoother convergence compared to DCGAN.

7 Conclusion

While DCGAN with Spectral Normalization offers a computationally efficient and bounded approach to stabilization, the WGAN-GP framework is generally superior for complex generative tasks. The guarantees of the Wasserstein Distance theory assure that the gradients of the Critic will be non-trivial even in cases where the output of the Generator is very wrong. Additionally, the Gradient Penalty completely avoids the problem of pathological weight clipping that is present in the original WGAN, which leads to reliable convergence, a high correlation between the loss of the Critic and the quality of the generated images, as well as greatly reduces the possibility of mode collapsing. While the theoretical complexity of the WGAN-GP is higher compared to the original, its practical application shows that the metrics like FID totally define the overall computational time.

References

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